An SDP Formula for the γ_2 Norm on $\ell_\infty^n \otimes \ell_2^m$

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Status

Claimed full answer, likely valid. The source paper asks, in Section 3 before introducing the mixed norm, whether the norm γ_2 on $\ell_\infty^n \otimes \ell_2^m$ can be computed easily as a standard convex optimization problem. The answer is yes: it has a polynomial-size semidefinite programming formula.

Source Question

The target is the following passage from Lee–Srinivasa–Junge–Romberg, *Approximately low-rank recovery from noisy and local measurements by convex program*, arXiv:2110.15205, Section 3:

> Although Lemma 3.1 also applies to $\ell_\infty^n \otimes \ell_2^m$, it has not been known whether the γ_2 norm on $\ell_\infty^n \otimes \ell_2^m$ can be computed easily as a standard convex optimization problem.

The paper then uses a different tensor norm, the mixed norm, because that norm is shown there to have an SDP formulation. The theorem below gives the missing SDP formulation for γ_2 itself.

Definitions

Let $M \in \mathbb{R}^{m\times n}$ be viewed as an operator $M : \ell_1^n \to \ell_2^m$, equivalently as an element of $\ell_\infty^n \otimes \ell_2^m$. Its Hilbert factorization norm is

$$\gamma_2(M) = \inf\left\{ \|A\|_{2\to 2}\, \|B\|_{\ell_1^n\to\ell_2^d} : d \in \mathbb{N},\ A : \ell_2^d \to \ell_2^m,\ B : \ell_1^n \to \ell_2^d,\ M = AB \right\}.$$

If $B = [b_1\ \cdots\ b_n]$, then

$$\|B\|_{\ell_1^n\to\ell_2^d} = \max_{1\le j\le n} \|b_j\|_2.$$

Proof Intuition

The usual max-norm SDP for $\ell_\infty^n \otimes \ell_\infty^m$ is just a Gram matrix encoding of two Hilbert families. In the present asymmetric case the right-hand factor is still controlled column-by-column, but the left factor is an operator into Euclidean space. Thus the right Gram matrix only needs bounded diagonal entries, while the left Gram matrix must be bounded in Loewner order by tI_m. Scaling the two factors balances their two bounds at the same value t, turning the product norm into a linear SDP objective.

Theorem

Theorem 1. *For every real matrix $M \in \mathbb{R}^{m\times n}$,*

$$\gamma_2(M) = \min t$$

where the minimum is taken over $t \geq 0$ and symmetric matrices $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{n\times n}$ satisfying

$$\begin{pmatrix} P & M \\ M^T & Q \end{pmatrix} \succeq 0, \qquad P \preceq tI_m, \qquad Q_{jj} \leq t \quad (1 \leq j \leq n).$$

Equivalently,

$$\gamma_2(M) = \inf\left\{\max\{\lambda_{\max}(P), \max_j Q_{jj}\} : \begin{pmatrix} P & M \\ M^T & Q \end{pmatrix} \succeq 0\right\}.$$

In particular γ_2 on $\ell_\infty^n \otimes \ell_2^m$ is computable by a standard semidefinite program.

Proof. First suppose $M = AB$, where $A : \ell_2^d \to \ell_2^m$ and $B : \ell_1^n \to \ell_2^d$. Write $b_j = Be_j$. Put $a = \|A\|_{2\to 2}$ and $b = \max_j \|b_j\|_2$. If $M \neq 0$, rescale the factorization by replacing A with $\widetilde{A} = (b/a)^{1/2}A$ and B with $\widetilde{B} = (a/b)^{1/2}B$; if $a = 0$ or $b = 0$, then $M = 0$ and the formula is trivial. The rescaled factorization still satisfies $M = \widetilde{A}\widetilde{B}$, and

$$\|\widetilde{A}\|_{2\to 2}^2 = ab, \qquad \max_j \|\widetilde{B}e_j\|_2^2 = ab.$$

Set

$$P = \widetilde{A}\widetilde{A}^T, \qquad Q = \widetilde{B}^T\widetilde{B}.$$

Then

$$\begin{pmatrix} P & M \\ M^T & Q \end{pmatrix} = \begin{pmatrix} \widetilde{A} \\ \widetilde{B}^T \end{pmatrix} \begin{pmatrix} \widetilde{A}^T & \widetilde{B} \end{pmatrix} \succeq 0,$$

while $P \preceq ab\, I_m$ and $Q_{jj} \leq ab$ for all j. Taking the infimum over all factorizations gives that the SDP value is at most $\gamma_2(M)$.

Conversely, suppose that P, Q, t are feasible. Since the block matrix is positive semidefinite, it is the Gram matrix of vectors $x_1, \ldots, x_m, y_1, \ldots, y_n$ in some finite-dimensional Hilbert space H:

$$P_{ik} = \langle x_i, x_k\rangle, \qquad M_{ij} = \langle x_i, y_j\rangle, \qquad Q_{jl} = \langle y_j, y_l\rangle.$$

Define $A : H \to \ell_2^m$ by $Ah = (\langle h, x_i\rangle)_{i=1}^m$ and define $B : \ell_1^n \to H$ by $Be_j = y_j$. Then $AB = M$. Moreover

$$\|A\|_{2\to 2}^2 = \|AA^*\|_{2\to 2} = \|P\|_{2\to 2} \leq t$$

and

$$\|B\|_{\ell_1^n \to H} = \max_j \|y_j\| = \max_j Q_{jj}^{1/2} \leq t^{1/2}.$$

Thus $\gamma_2(M) \leq t$. Taking the infimum over feasible triples gives the reverse inequality. $\square$

Verification Notes

The proof is finite-dimensional and uses only the standard Gram factorization criterion for positive semidefinite block matrices. The SDP has variables t, P, and Q and uses only linear matrix inequalities:

$$\begin{pmatrix} P & M \\ M^T & Q \end{pmatrix} \succeq 0, \qquad tI_m - P \succeq 0, \qquad Q_{jj} \leq t.$$

Hence it is a standard semidefinite program in the usual sense.

Remark 1. *The formula differs from the mixed-norm SDP in the source paper exactly where the left Hilbert factor is measured. The mixed norm uses a Frobenius/trace bound on the left Gram matrix; γ_2 uses the spectral/Loewner bound $P \preceq tI_m$.*

Limitations and Novelty Check

This packet answers only the computability question for the γ_2 norm. It does not claim that substituting this norm into the source paper's estimator automatically preserves the same statistical error analysis; that would require fresh entropy estimates for the corresponding SDP ball.

Cheap run-index searches found no prior packet for arXiv:2110.15205 or for the phrases "γ_2", "$\ell_\infty^n \otimes \ell_2^m$", and semidefinite computability. Web searches on 2026-06-28 for `gamma_2 ell_infty ell_2 semidefinite programming`, `gamma_2 norm SDP ell_2 ell_infty`, and related factorization-norm queries found the standard max-norm/$\ell_\infty \otimes \ell_\infty$ SDP but no explicit formula for this asymmetric $\ell_\infty \otimes \ell_2$ case.

Human Review Recommendation

Review as a short full solution to the source paper's explicit computability question. The main points to check are the orientation convention for $\ell_\infty^n \otimes \ell_2^m$ as operators $\ell_1^n \to \ell_2^m$, and the balanced rescaling step converting the factorization product into one SDP scalar t.

References

[1] K. Lee, R. S. Srinivasa, M. Junge, and J. Romberg, *Approximately low-rank recovery from noisy and local measurements by convex program*, arXiv:2110.15205.

[2] N. Srebro, J. D. M. Rennie, and T. S. Jaakkola, *Maximum-margin matrix factorization*, Advances in Neural Information Processing Systems 17, 2004.